Beginning Hypothesis Testing

October 28–30, 2026 EN.553.211 Zan Ahmad

Chapter 7 used samples to estimate unknown population parameters. Chapter 8 asks whether a particular claim about a parameter is compatible with the observed data. These notes develop the reference model and interpretation of a test, then connect them to confidence intervals through Friday practice. Read the explanations on pp. 1–3; work through the questions on pp. 4–8. The optional continuation on p. 9 introduces critical values and is not required for the Friday questions.

Learning goals

By the end of this introduction, you should be able to:

- Translate a research question into H_0 and H_a before calculating.
- Explain how the null hypothesis supplies a reference distribution, and distinguish a random statistic from its observed value.
- Compute and interpret a large-sample mean-test p-value, choose the correct tail, and state a conclusion at a specified significance level.

1. From an interval to a question about a claim

Recall Chapter 7 Problem 3. When p was unknown, we estimated it and constructed a confidence interval. In the separate model with $p = 0.75$, we could calculate how often a sample proportion would be at most 0.70. Hypothesis testing uses this second kind of reasoning: *temporarily assume a claim, then ask how unusual the observed result would be under that claim.*

A population claim H_0 → Predicted sampling behavior under H_0 → Compare the observed result with that behavior

The claim supplies the reference model. The data supply the observed result. A small tail probability indicates tension between them, subject to the sampling assumptions.

2. State the hypotheses

The **null hypothesis** H_0 is the claim used to calibrate the test. For a population mean we write $H_0 : \mu = \mu_0$, where μ_0 is the specified **null value**. The **alternative hypothesis** H_a expresses the research question:

Question	Alternative	Test direction
Has the mean increased?	$H_a : \mu > \mu_0$	Right-sided
Has the mean decreased?	$H_a : \mu < \mu_0$	Left-sided
Has the mean changed?	$H_a : \mu \neq \mu_0$	Two-sided

Choose the direction from the scientific question, before seeing the result. The slides compare this to a court's presumption of innocence: insufficient evidence to reject a claim does not prove the claim true. The analogy describes the burden of evidence, not a numerical legal standard.

Check for understanding. A treatment is intended to *reduce* mean recovery time from 12 days. State H_0 and H_a. Would a sample mean above 12 favor this alternative?

The Null Distribution

3. Measure a discrepancy in standard errors

Let $X_1, \ldots, X_n$ be independent, identically distributed observations with mean μ and finite variance $0 < \sigma^2 < \infty$. Recall

$$\overline{X} = \frac{1}{n}\sum_{i=1}^{n} X_i, \qquad \mathrm{E}[\overline{X}] = \mu, \qquad \mathrm{Var}(\overline{X}) = \frac{\sigma^2}{n}, \qquad \mathrm{SE}(\overline{X}) = \frac{\sigma}{\sqrt{n}}.$$

For a normal population, $\overline{X} \sim N(\mu, \sigma/\sqrt{n})$ exactly. As in our Chapter 7 notes, the second argument of N denotes *standard deviation*. Standardizing gives

$$\frac{\overline{X} - \mu}{\sigma/\sqrt{n}} \sim N(0,1).$$

Now impose the null claim. If $H_0 : \mu = \mu_0$ is true and σ is known, the test statistic is

$$Z = \frac{\overline{X} - \mu_0}{\sigma/\sqrt{n}} \sim N(0,1) \qquad \text{under } H_0.$$

For a nonnormal population, the normal statement is an approximation justified by an adequate CLT approximation. When σ is unknown, use

$$S^2 = \frac{1}{n-1}\sum_{i=1}^{n}(X_i - \overline{X})^2, \qquad \frac{\overline{X} - \mu_0}{S/\sqrt{n}} \overset{\text{approx}}{\sim} N(0,1) \qquad \text{under } H_0.$$

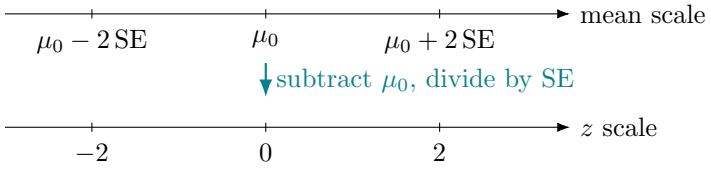
The course uses $n \geq 30$ as a large-sample guideline. It is not a guarantee for every population shape; substantial skewness or heavy tails may need larger samples. A large sample does not repair dependence or biased sampling. Small-sample methods come later.

4. Keep the random quantity and the observation separate

Before sampling, $\overline{X}$ and S vary. After sampling, $\overline{x}$ and s are fixed numbers. With unknown σ, the **observed test statistic** is

$$z = \frac{\overline{x} - \mu_0}{s/\sqrt{n}} = \frac{\text{observed estimate} - \text{null value}}{\text{estimated standard error}}.$$

The numerator is a discrepancy in the original units. Dividing by the standard error expresses that discrepancy in units of sampling variability. Thus $z = 2$ means the estimate lies two estimated standard errors above the null value.



Why use a tail? In a continuous normal model, $\Pr(\overline{X} = \mu_0) = 0$, even when H_0 is true. Exact equality is therefore the wrong criterion. We ask for results *at least as extreme*, in the direction specified by H_a.

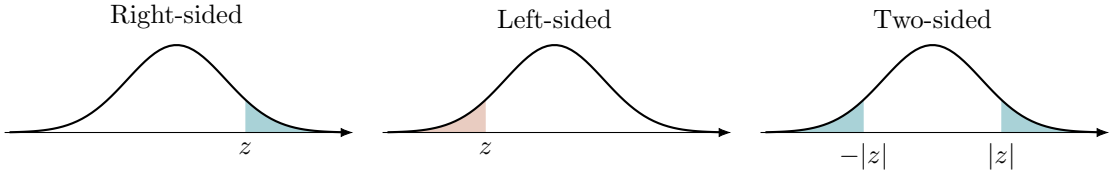
Check for understanding. Suppose the discrepancy points toward H_a. If $\overline{x} - \mu_0$ stays fixed but the standard error doubles, what happens to z? Explain why the same raw difference now provides weaker evidence.

What Counts as More Extreme

5. The alternative determines the shaded area

The p-**value** is the probability, *assuming* H_0 *and the sampling model*, of a test statistic at least as favorable to H_a as the observed statistic. For a standard-normal reference and observed value z:

Alternative	Outcomes at least as extreme	p-value
$\mu > \mu_0$	$Z \geq z$	$1 - \Phi(z)$
$\mu < \mu_0$	$Z \leq z$	$\Phi(z)$
$\mu \neq \mu_0$	$Z \leq -\lvert z\rvert$ or $Z \geq \lvert z\rvert$	$2[1 - \Phi(\lvert z\rvert)]$



The shaded tails illustrate the three rules; these are schematic cutoffs, not a new data set.

Why the factor of two? For a two-sided question, equally large discrepancies in either direction count against H_0. Symmetry of the standard normal gives

$$\Pr(|Z| \geq |z|) = \Pr(Z \leq -|z|) + \Pr(Z \geq |z|) = 2\Pr(Z \geq |z|).$$

A one-sided p-value is not always half a two-sided one: the observed discrepancy must point toward the specified alternative. For example, a large positive z yields a large *left-sided* p-value.

6. Two decisions, carefully worded

$$\begin{aligned} p\text{-value} \leq \alpha &\implies \text{reject } H_0; \\ p\text{-value} > \alpha &\implies \text{fail to reject } H_0. \end{aligned}$$

Failure to reject means **insufficient evidence for the specified alternative at this level**. It does not establish equality. Statistical significance also does not measure the scientific importance of a difference; report the observed effect and its context.

Explain before calculating

For each practice question, identify the population parameter, specify H_0 and H_a, state the reference model and its assumptions, and distinguish the statistic from its observed value. Then connect the probability statement to the research question and conclude in context.

Check for understanding. A test yields a p-value of 0.08 with $\alpha = 0.05$. State the decision and explain why it does not prove H_0.

Friday Practice Chapter 7 Review

Fall-break problem 3. An unknown rate and a specified probability

A monitoring program calls a returned water sample *usable* if it meets a prespecified handling checklist. In a random sample of 240 returned kits, 168 samples are usable. Let p be the population proportion of usable samples among this program's returned kits. Assume independent Bernoulli outcomes with a common probability p, and a population much larger than the sample. These data are illustrative.

Use $z_{0.025} = 1.96$ and let Φ denote the standard normal CDF.

(a) Estimate p and check the observed success and failure counts. Construct an approximate 95% confidence interval for p, showing the estimated standard error and margin of error. Interpret the endpoints as percentages and identify the population they describe.

(b) Now consider a *separate hypothetical model* in which the population proportion is known to be $p = 0.75$. For a new independent sample of $n = 240$, find the approximate probability $\Pr(\widehat{p} \leq 0.70)$. Check the expected counts, standardize, and use a direct normal approximation without continuity correction.

(c) Explain why the standard errors in parts (a) and (b) use different proportions. Which question estimates a fixed unknown parameter, and which assigns a probability to a random sample result?

(d) *Think ahead using the sample-size planning section.* A new program wants a planned 95% normal-interval margin of error of at most 4 percentage points. No reliable advance estimate of its population proportion is available. Find the smallest integer sample size that meets this target under the worst-case choice of p. Justify that choice and state whether your answer is a required *total* or a number to add to 240.

Problem 1 The GPA Example

Hypotheses and the statistic

Use the example from the recorded Chapter 8 lecture. The historical reference mean GPA is 3.37. An independent random sample of 50 current students has $\overline{x} = 3.47$ and $s = .31$. Before collecting the sample, the research question was whether the population mean GPA had increased. The chosen significance level is $\alpha = .05$.

Use the lecture's large-sample normal approximation, assuming its sampling conditions are adequate. The population standard deviation is unknown. Write Φ for the standard normal CDF.

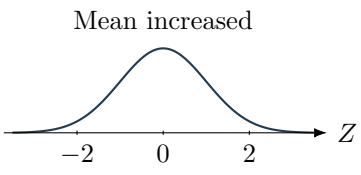
(a) **Hypotheses and parameters.** Define the population parameter μ. Explain the roles of the *null hypothesis* and *alternative hypothesis*, and state both for this question. Identify the *null value* μ_0. Is a hypothesis a statement about μ or about the observed $\overline{x}$?

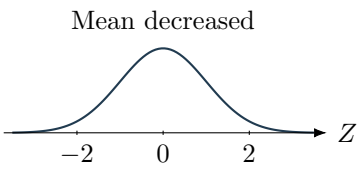
(b) **Statistic and reference distribution.** Write the test statistic in terms of the random quantities $\overline{X}$ and S, then calculate its observed value z using $\overline{x}$ and s. Explain what the numerator and denominator measure. Define the *null distribution*; state the distribution used here and the assumption under which it applies. Which is random before sampling: the statistic or the null value?

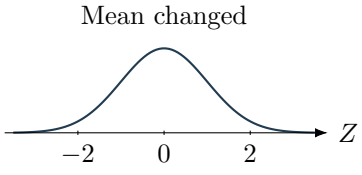
Problem 1 Continued

Data reference: $n = 50$, $\overline{x} = 3.47$, $s = .31$; historical mean 3.37.

(c) **The p-value and its tail.** Define a p-value in words, including both what is assumed and what counts as "at least as favorable." For each separately specified research question below, state H_a, write a probability involving Z and the observed z, shade its region, and calculate the p-value. The data stay the same; these are three alternative questions, each chosen *before* seeing data.







Mark the observed boundary or boundaries and shade the relevant region. Use a normal table or calculator.

(d) **Significance and conclusion.** Define the significance level as used so far in this lecture. For the original *increase* question, state the decision at .05 and the conclusion about the population. Reassess the same evidence using a prechosen .01 level. Which of the data, z, p-value, threshold, and decision change?

(e) **Interpretation check.** Explain what is wrong with each statement and replace it with a correct one: (i) "The p-value is the probability that our null hypothesis is wrong." (ii) "If we fail to reject, we have proved that the population mean is 3.37."

Problem 2 The Chapter 7 Connection

One sample and two research questions

Return to Chapter 7 Problem 3. In a random sample of 240 returned water-sample kits, 168 are usable. Let p be the population proportion of usable samples *among this program's returned kits*. Assume independent Bernoulli outcomes with a common p and a population much larger than the sample. These data are illustrative.

New question. Before sampling, the program asks whether the usable-sample proportion *differs from* .75. The decline comparison below is a second, separately preplanned question. Use $\alpha = .05$.

(a) **Retrieve the confidence interval.** Use your Chapter 7 work, or recompute the estimate $\widehat{p}$, estimated standard error, and approximate 95% confidence interval. Use $z_{.025} = 1.96$. Does the interval include .75? State what population quantity the interval estimates.

(b) **Build a null reference model.** State H_0 and H_a for the new question. Check the expected success and failure counts under H_0. The model from Chapter 7 Problem 3(b) supplies

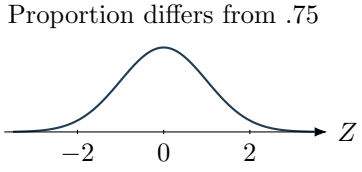
$$Z = \frac{\widehat{p} - .75}{\sqrt{.75(.25)/240}} \overset{\text{approx}}{\sim} N(0,1) \quad \text{when } p = .75.$$

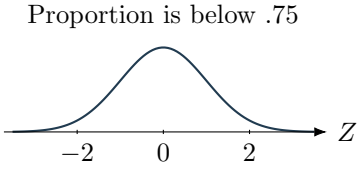
Use this supplied model to calculate the observed z. Explain why the interval in (a) uses an estimated standard error but this reference calculation uses .75. In each calculation, identify what is fixed and what varies across repeated samples.

Problem 2 Continued

Data reference: 168 usable samples among 240 returned kits; reference proportion .75; $\alpha = .05$. Reuse your calculations from the preceding page.

(c) **Turn the probability into evidence.** For the *differs* question, mark the observed boundaries and shade the appropriate region on the first curve. Calculate the p-value using the direct normal approximation, without continuity correction. Give a decision and a contextual conclusion. Does this decision agree with the fact you observed about the 95% interval?





(d) **Compare a planned decline question.** A second analyst had specified $H_a : p < .75$ *before* seeing the same data. Shade that analyst's region on the second curve. Use your earlier calculation to obtain the p-value and decision at .05. Explain how the two analysts can reach different decisions without changing the data or the null distribution.

(e) **Reconcile the interval and the tests.** A classmate says, "A two-sided 95% confidence interval decides every 5% test of its null value." Use your results to assess the claim. Identify both the difference in tail questions and the difference between the two standard-error formulas. Should an analyst choose the direction of H_a after seeing which tail gives the smaller p-value? Explain.

Optional Extension: Critical Values

Read after the core lesson. This page develops the rejection-region material in slides 21–23. It is not needed to complete the core example by its p-value.

7. A decision rule has an error rate

A **Type I error** is rejecting H_0 when H_0 is true. For an exactly calibrated continuous test of size α,

$$\Pr(\text{reject } H_0 \mid H_0 \text{ is true}) = \alpha.$$

For our large-sample tests, the calibration is approximate. A significance level of .05 describes a long-run false-rejection rate under the null model; it is not the probability that this particular conclusion is wrong.

8. Convert a tail threshold to a statistic threshold

The course uses **upper-tail** critical values: $\Pr(Z \geq z_\alpha) = \alpha$. For a right-sided test, monotonicity of Φ gives

$$1 - \Phi(z) \leq \alpha \quad \Longleftrightarrow \quad \Phi(z) \geq 1 - \alpha \quad \Longleftrightarrow \quad z \geq z_\alpha.$$

The **rejection region** R_0 is the set of statistic values for which we reject H_0:

Alternative	Rejection region R_0
$\mu > \mu_0$	$[z_\alpha, \infty)$
$\mu < \mu_0$	$(-\infty, -z_\alpha]$
$\mu \neq \mu_0$	$(-\infty, -z_{\alpha/2}] \cup [z_{\alpha/2}, \infty)$

For a two-sided test, split the total null rejection probability α equally between the two tails. This is the same critical-value convention used for Chapter 7 confidence intervals.

9. Connect the two descriptions

The rejection threshold is fixed before sampling. The observed statistic and its p-value depend on the sample. The critical-value method compares a statistic with a fixed boundary; the p-value method compares the corresponding tail probability with α.

Check for understanding. Return to the GPA example in Problem 1. For $\alpha = .05$, use $z_{.05} \approx 1.645$ to specify the rejection region. Compare your observed statistic with it. Does this agree with your p-value decision?

Check for understanding. Why does a two-sided 5% test use 1.96 rather than 1.645?

Chapter 7 Problem 3

Question on original notes page 4.

(a) Estimate an unknown population proportion

The observed proportion is $\widehat{p} = 168/240 = 0.70$. The observed success and failure counts, 168 and 72, both exceed the course guideline of 5. The sampling assumptions are also needed; this count check alone does not repair dependence or selection bias.

$$\widehat{\mathrm{SE}}(\widehat{p}) = \sqrt{\frac{(0.70)(0.30)}{240}} \approx 0.0295804, \qquad ME = 1.96\widehat{\mathrm{SE}} \approx 0.0579776.$$

$$\boxed{CI_{95\%}(p) = 0.70 \pm 0.0579776 = [0.64202,\ 0.75798].}$$

We are approximately 95% confident that **64.20% to 75.80%** of this program's population of returned kits yield usable samples. The target is returned kits, not all kits distributed or all households. The margin is about 5.80 percentage points. About 95% of intervals from repeating this sampling procedure would contain the fixed p.

(b) A probability under a separate specified model

Here the model states $p = 0.75$. For a new sample, $X \sim \mathrm{Binomial}(240, 0.75)$ and $\widehat{p} = X/240$. The expected success and failure counts are 180 and 60, both greater than 5. Use the specified p in the sampling distribution:

$$\mathbb{E}(\widehat{p}) = 0.75, \qquad \mathrm{SE}(\widehat{p}) = \sqrt{\frac{(0.75)(0.25)}{240}} \approx 0.0279508.$$

With the requested direct normal approximation,

$$\Pr(\widehat{p} \le 0.70) \approx \Phi\left(\frac{0.70 - 0.75}{0.0279508}\right) = \Phi(-1.78885) \approx \boxed{0.0368}.$$

The underlying event is $X \le 168$. No continuity correction was used, as requested. The approximation describes a roughly 3.68% chance of this sample event under the stipulated model.

(c) Why the two standard errors differ

In part (a), p is fixed but unknown; $\widehat{p} = 0.70$ estimates it, so $\sqrt{\widehat{p}(1-\widehat{p})/n}$ estimates the SE for a CI. In part (b), $p = 0.75$ is given by the hypothetical model, so use $\sqrt{p(1-p)/n}$ to calculate a probability for a random sample proportion. One procedure estimates a parameter; the other predicts sample variation under a specified parameter value.

(d) Conservative sample-size planning

Four percentage points means $\varepsilon = 0.04$. Since $p(1-p) = \frac{1}{4} - (p - \frac{1}{2})^2 \le \frac{1}{4}$, the largest planned SE occurs at $p = 0.5$. Therefore require

$$1.96\sqrt{\frac{1/4}{n}} \le 0.04 \quad \Longleftrightarrow \quad n \ge \frac{1.96^2(1/4)}{0.04^2} = 600.25, \qquad \boxed{n = 601}.$$

Round *up*. This is a required **total** from the new program, not a number added to 240. It controls the planned normal-theory margin for every p; it does not guarantee exact 95% coverage where the normal approximation is poor, particularly near 0 or 1.

Chapter 8 Problem 1

(a) A claim about a population parameter

μ is the mean GPA of the current student population from which the random sample is drawn. The null hypothesis supplies the reference claim $H_0 : \mu = 3.37$. The research question supplies $H_a : \mu > 3.37$. The null value is the fixed number $\mu_0 = 3.37$. Hypotheses concern μ; $\overline{x} = 3.47$ is an observed estimate, not the parameter.

(b) Random statistic and observed value

$$Z = \frac{\overline{X} - 3.37}{S/\sqrt{50}} \overset{\text{approx}}{\sim} N(0,1) \text{ under } H_0, \qquad z = \frac{3.47 - 3.37}{.31/\sqrt{50}} \approx 2.28099.$$

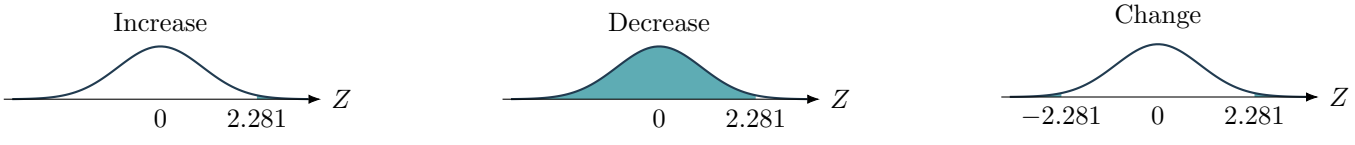
The numerator is the sample mean's discrepancy from the null value. The estimated sampling SE is $.31/\sqrt{50} \approx .0438406$ GPA units. The observed mean is about 2.281 SEs above the null value. Before sampling, $\overline{X}$, S, and Z vary; the stipulated null value does not. After sampling, $\overline{x}$, s, and z are fixed observations.

The *null distribution* is the reference distribution of the statistic when H_0 and the sampling assumptions hold. Here it is approximately standard normal under the lecture's large-sample method. Substituting an estimated SD does not make this an exact standard-normal result. The sample-size rule is a course convention, not a universal guarantee for arbitrary populations.

(c) Evidence depends on the alternative

A p-value is the probability, assuming H_0 and the model, of a statistic at least as favorable to the *specified* H_a as the observed statistic. With $z = 2.28099$:

Alternative	Tail probability	Approximate p-value
$\mu > 3.37$	$P(Z \geq z) = 1 - \Phi(z)$	.01127
$\mu < 3.37$	$P(Z \leq z) = \Phi(z)$	.98873
$\mu \neq 3.37$	$P(\lvert Z\rvert \geq \lvert z\rvert) = 2[1 - \Phi(\lvert z\rvert)]$	.02255



(d) Thresholds and conclusions

So far, α is the *prechosen threshold* for the p-value. Reject when p-value$\leq \alpha$; otherwise fail to reject. For the increase question, $.01127 < .05$: there is evidence of an increased population mean GPA. At .01, fail to reject: the evidence does not meet that stricter threshold. The data, observed z, and p-value stay the same. Only the threshold and decision change.

(e) Repair the interpretations

(i) Under the null model, about 1.13% of samples would yield a standardized increase at least this large. The p-value is not the probability that H_0 is wrong (or true). The null model is assumed when this tail probability is calculated.

(ii) Failure to reject means insufficient evidence for the specified alternative at that level; it does not prove equality.

Chapter 8 Problem 2

(a) Retrieve the original estimation calculation

$$\widehat{p} = \frac{168}{240} = .70, \qquad \widehat{\mathrm{SE}} = \sqrt{\frac{.70(.30)}{240}} \approx .0295804.$$

The observed success/failure counts are 168 and 72. The 95% normal interval is

$$.70 \pm 1.96(.0295804) = [.6420224, .7579776].$$

It includes .75. It estimates the population usable-sample proportion among this program's *returned kits*, about 64.20% to 75.80%. It is not an interval for the probability that the fixed parameter lies inside the realized endpoints.

(b) Condition on the null claim

For the new question, $H_0 : p = .75$ and $H_a : p \neq .75$. The expected counts under H_0 are $240(.75) = 180$ and $240(.25) = 60$, meeting the course's normal-approximation check. Use the supplied reference model:

$$\mathrm{SE}_0 = \sqrt{\frac{.75(.25)}{240}} \approx .02795085, \qquad z = \frac{.70 - .75}{.02795085} \approx -1.788854.$$

For estimation, the unknown fixed p is estimated by the random $\widehat{p}$, giving a plug-in SE. Across repeated samples both interval center and estimated SE vary. For the null reference calculation, $p = .75$ and its SE are fixed; the sample count and $\widehat{p}$ vary. The observed .70 is fixed after collection. The symbol p for the proportion is distinct from the phrase p-value.

(c) The two-sided question

$$p\text{-value} \approx P(|Z| \geq 1.788854) = 2\Phi(-1.788854) \approx .0736383.$$

Shade both tails beyond -1.789 and $+1.789$. Since $.07364 > .05$, fail to reject. There is insufficient evidence at 5% that the population usable proportion differs from .75. This agrees with the observed inclusion of .75 in the 95% interval for these data. It does not prove exact equivalence of the two procedures for every sample.

(d) The planned lower-tail question

For $H_a : p < .75$, shade only $Z \leq -1.788854$. Original Problem 3(b)'s probability becomes the one-sided p-value:

$$p\text{-value} \approx \Phi(-1.788854) \approx .0368191 < .05.$$

Reject: there is evidence of a decline below .75 for this preplanned directional question. The null model, observed z, and data are identical; the tail event differs. The two-sided event includes equally large deviations in *both* directions and has twice the area here. Rounding z to -1.79 gives table values .0367 and .0734.

(e) Why the interval does not decide every test

A two-sided 95% interval addresses both directions; it cannot automatically determine a one-sided 5% test. Also, this Chapter 7 Wald interval uses $\widehat{p}$ in its SE, whereas this null test uses $p_0 = .75$. Thus even the two-sided comparison is not an exact inversion of *this* interval method. Agreement in (c) is a result for these data. A 95% proportion interval and a 5% test need not agree when their constructions differ.

The alternative represents the research question chosen before inspecting the sample. Choosing the smaller tail after seeing the result changes the procedure and does not justify the planned one-sided interpretation.

Checks for Understanding

Answers to the short checks on original notes pages 1–3 and 9.

Page 1: a treatment intended to reduce recovery time

The population parameter is the mean recovery time μ under the treatment. Use $H_0 : \mu = 12$ days and $H_a : \mu < 12$ days. A sample mean above 12 points away from this alternative and does not support a reduction. Choose the alternative before observing the sample.

Page 2: the same discrepancy with twice the SE

Since $z = (\overline{x} - \mu_0)/SE$, doubling the denominator halves z when the numerator is fixed. Its magnitude moves toward zero. A discrepancy in the direction of H_a now represents less unusual sampling variation under the null, so its corresponding p-value increases. The raw difference is unchanged; its size relative to sampling uncertainty is smaller.

Page 3: a p-value of .08 at the .05 level

Because $.08 > .05$, fail to reject H_0. The data do not provide enough evidence for the specified alternative at this significance level. This does not prove the null; a sample can fail to detect an effect that is present.

Page 9: the GPA rejection region

For the prechosen increase question, $H_a : \mu > 3.37$ and $\alpha = .05$. The upper-tail convention gives

$$R_0 = [z_{.05}, \infty) \approx [1.645, \infty).$$

The observed $z \approx 2.281$ belongs to this region. Reject H_0, agreeing with the upper-tail p-value $.01127 < .05$. Both methods express the same decision rule under the same reference model.

Page 9: why a two-sided 5 percent test uses 1.96

A two-sided test divides the total null rejection probability .05 between two tails, .025 each. Thus $P(Z > 1.96) \approx .025$, and the rejection region is

$$(-\infty, -1.96] \cup [1.96, \infty).$$

Using 1.645 in both tails would allocate about .05 to each tail, for total size about .10, rather than .05.

Read the conclusions in context. The same observed data can address different preplanned research questions. Changing a threshold does not change the data or the p-value. Choosing a direction after inspecting which tail is favorable changes the testing procedure.

Problem 2(c)–(d): the tail regions



From Evidence to a Testing Procedure

Chapter 8 continuation November 1, 2026 EN.553.211

The first part of these notes built hypotheses, null sampling distributions and p-values. We now use that same reasoning to design a decision rule, examine its mistakes, and compare means and proportions. The central question remains: *How much sampling variation should we expect under a specified model?*

Reading map

Earlier material, preserved in its original order	Pages
Hypotheses, null distributions and p-values	1–3
Friday questions and an introductory critical-value page	4–9
Friday worked solutions	10–13
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The earlier pages retain their original page references. The continuation begins where the October 28 recording stopped: original slide 21, *Significance level & critical values*. Posting a topic here does not imply that we have already discussed it in class.

Learning goals for the continuation

- Translate a significance level into a rejection region and relate that region to a p-value.
- Describe Type I and Type II errors in context, and compute power under a specified alternative model.
- Derive the appropriate standard error for one mean, independent means, paired differences, one proportion and two proportions.
- Explain the connection to confidence intervals, including when a confidence interval and a test use different standard errors.

One structure, several sampling models

$$\text{test statistic} = \frac{\text{estimate} - \text{null value}}{\text{standard error appropriate to the null model}}$$

The numerator names the discrepancy. The denominator sets its scale. The alternative tells us which discrepancies count as evidence. Changing the parameter changes the estimate and its standard error; it does not change this logic.

Notation follows the course: $N(\mu, \sigma)$ lists the **standard deviation** as its second argument. Write $\Phi(z) = \Pr(Z \le z)$ for $Z \sim N(0,1)$. Normal approximations below assume an appropriate independent sampling design; a large n cannot fix biased sampling or dependence.

Critical Values and Rejection Regions

A p-value describes the evidence from the observed sample. A significance level α sets a decision rule *before* sampling: reject H_0 when the p-value is at most α.

Turn a tail probability into a cutoff

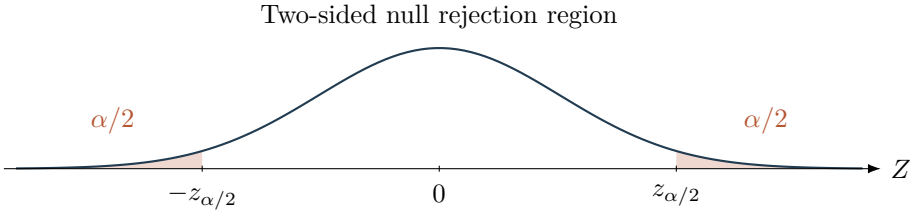
The course writes z_a for the upper-tail standard-normal quantile:

$$\Pr(Z \geq z_a) = a, \qquad z_a = \Phi^{-1}(1-a).$$

For example, $z_{.05} \approx 1.645$, $z_{.025} \approx 1.960$ and $z_{.01} \approx 2.326$. For an approximately standard-normal statistic under H_0:

Alternative	p-value for observed z	Reject when
Parameter $>$ null value	$1 - \Phi(z)$	$z \geq z_\alpha$
Parameter $<$ null value	$\Phi(z)$	$z \leq -z_\alpha$
Parameter $\neq$ null value	$2[1 - \Phi(\lvert z \rvert)]$	$\lvert z \rvert \geq z_{\alpha/2}$

In the two-sided test, split the total null tail probability α equally between both directions. Using z_α on both sides would give total null rejection probability 2α.



The middle region is the **nonrejection region**. Landing there means that the procedure has not found sufficient evidence against H_0. It does not establish that H_0 is true.

A reusable testing workflow

State the population parameter and H_0, H_a; identify the sampling design; check the model and normal approximation; compute the statistic; obtain the correctly directed p-value or compare with the critical region; conclude in the original context. Report the effect estimate as well as the decision.

Pause and explain. At $\alpha = .05$, is $z = 1.8$ significant for a right-sided test? For a two-sided test? Why can the answers differ?

Carpenters: A Complete Mean Test

Lecture-slide example: Carpentry, original slide 31 / slide-deck PDF p. 41.

Question. You wish to show that the average hourly wage of carpenters in a particular city is different from \$19, the city average. You survey a sample of 100 carpenters and find the sample average of \$20 with a sample standard deviation of \$6. Use a significance level of $\alpha = 0.05$ to test your hypothesis.

1. Name the parameter and model

Let μ be the population mean hourly wage of carpenters in this city. The claim is about a population mean, not about whether the observed sample mean equals 19:

$$H_0 : \mu = 19, \qquad H_a : \mu \neq 19.$$

The slide does not specify the selection mechanism. We proceed *assuming* the survey gives independent, representative observations with finite variance and an adequate CLT approximation. Since σ is unknown, this Chapter 8 calculation uses the large-sample plug-in standard error $s/\sqrt{n}$.

2. Measure the discrepancy and find its probability

$$\widehat{\mathrm{SE}}(\overline{X}) = \frac{6}{\sqrt{100}} = 0.6 \text{ dollars}, \qquad z = \frac{20 - 19}{0.6} = 1.6667.$$

The estimate is one dollar above the claim, or about 1.67 estimated standard errors above it. Because the question says *different*, both directions count:

$$p\text{-value} = 2[1 - \Phi(1.6667)] \approx 0.09558.$$

Since $0.09558 > 0.05$, **fail to reject** H_0. Equivalently, $|1.6667| < 1.960$, so the observation is outside the rejection region. There is insufficient evidence at the 5% level that the city's mean carpenter wage differs from \$19 per hour.

3. Connect directly to Chapter 7

Using the same approximate model and standard error, the 95% mean confidence interval is

$$20 \pm 1.960(0.6) = [18.824,\ 21.176].$$

It contains 19, agreeing with the two-sided 5% test. More generally,

$$\left|\frac{\overline{x} - \mu_0}{s/\sqrt{n}}\right| > z_{\alpha/2} \iff \mu_0 \notin [\overline{x} - z_{\alpha/2}s/\sqrt{n},\ \overline{x} + z_{\alpha/2}s/\sqrt{n}].$$

Apart from the convention at the exact critical boundary, this equivalence uses the *same* model, standard error and two-sided level in both procedures. The interval describes uncertainty about the mean; it is not an interval containing 95% of individual wages.

The sample suggests a \$1 increase. The test asks whether that discrepancy is large relative to uncertainty. Failure to reject does not erase the estimated difference or prove that the true difference is zero.

What Can a Test Get Wrong?

A decision depends on a random sample. Even when the sampling assumptions hold, repeated samples can lead to different decisions. Error probabilities describe the behavior of a *procedure* across such repetitions.

	H_0 is true	A specified H_a value is true
Reject H_0	Type I error	Correct detection
Fail to reject H_0	Correct nonrejection	Type II error

Type I error: signal when the null is true

For a simple null with a calibrated continuous test,

$$\alpha = \Pr(\text{reject } H_0 \mid H_0 \text{ true}).$$

In the carpenter example: conclude that the population mean wage differs from \$19 when it actually equals \$19. A test with $\alpha = .05$ is designed to make this error in 5% of repeated samples under the null model. With our estimated standard error and normal approximation, that calibration is approximate.

Type II error: miss a specified departure

For a possible true mean $\mu_1 \neq 19$, define

$$\beta(\mu_1, \alpha) = \Pr(\text{fail to reject } H_0 \mid \mu = \mu_1), \qquad \text{Power}(\mu_1) = 1 - \beta(\mu_1, \alpha).$$

In the carpenter example: fail to detect a difference when the mean really is, for instance, \$20. **Power** is the probability that the procedure detects that specified difference.

Why there is no single β for the whole alternative

The alternative $\mu \neq 19$ contains values very close to 19 and values far from it. These produce different sampling distributions and are not equally easy to distinguish from the null. To calculate a Type II error probability, specify a true alternative value and the variability model as well as the test.

These are conditional probabilities in the sampling direction. α is not $\Pr(H_0 \text{ true} \mid \text{rejection})$, and the p-value is not $\Pr(H_0 \text{ true} \mid \text{data})$. Power is not the probability that the observed conclusion is correct.

Pause and explain. If the carpenter mean truly equals 20 but our sample leads us to fail to reject 19, which error occurred? Could the observed sample alone tell us that this error occurred?

Carpenters: Keep the Rule, Change the World

Lecture-slide example: Carpentry power calculation, original slide 38 / PDF pp. 49–51.

Question. What is the power of the carpenter test if the population mean actually is \$20 per hour?

Planning approximation. As in the slides, treat 6 dollars as a fixed population standard deviation for this calculation. In the original test, 6 was the observed sample SD; freezing it is an additional approximation, not a claim that the unknown population SD has become known.

1. Determine the rule under the null

At $\alpha = .05$, fail to reject when

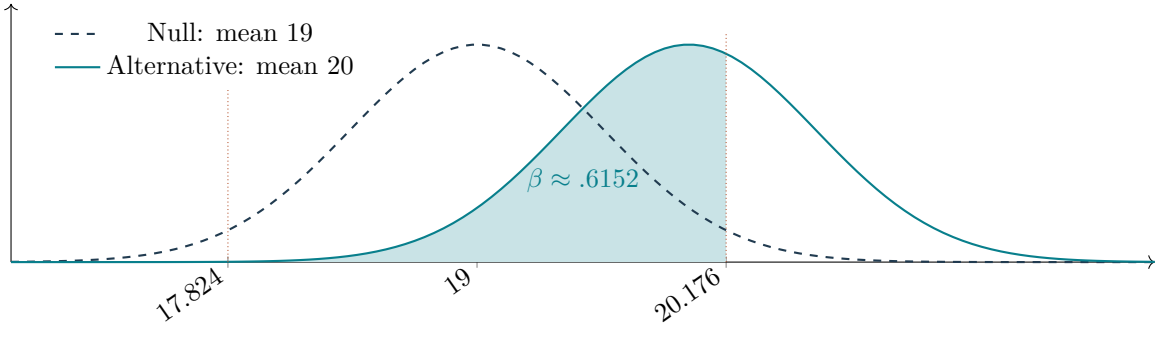
$$-1.960 \le \frac{\overline{X} - 19}{6/\sqrt{100}} \le 1.960 \iff 17.824 \le \overline{X} \le 20.176.$$

These cutoffs are centered at the **null mean 19**. The observed confidence interval on p. 16 was centered at the **sample mean 20**. They answer different questions.

2. Evaluate that unchanged rule under the alternative

If $\mu = 20$, the planning model gives $\overline{X} \overset{\text{approx}}{\sim} N(20, 0.6)$. Thus

$$\begin{aligned}
\beta(20, .05) &\approx \Pr(17.824 \le \overline{X} \le 20.176 \mid \mu = 20) \\
&= \Phi\left(\frac{20.176 - 20}{0.6}\right) - \Phi\left(\frac{17.824 - 20}{0.6}\right) \\
&= \Phi(0.2933) - \Phi(-3.6267) \approx 0.6152, \\
\text{Power}(20) &\approx 1 - 0.6152 = \boxed{0.3848}.
\end{aligned}$$



The shaded area under the alternative curve is the chance of missing a true mean of 20. The two unshaded tails under that *same* curve give power. A 5% Type I error rate does not imply a 95% chance of detecting every real difference.

Power, Precision and Study Design

The carpenter calculation is an instance of a general procedure: **fix the rejection region using H_0; then calculate its probability under a specified alternative.** Do not move the cutoffs to the alternative mean.

For a two-sided mean test with fixed σ, set

$$L = \mu_0 - z_{\alpha/2}\frac{\sigma}{\sqrt{n}}, \qquad U = \mu_0 + z_{\alpha/2}\frac{\sigma}{\sqrt{n}}.$$

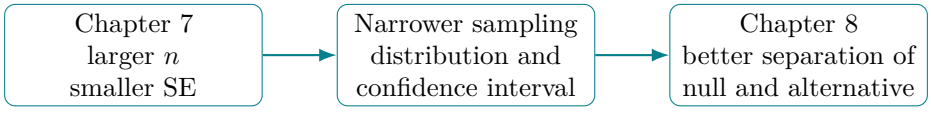
Under true mean μ_1,

$$\beta(\mu_1, \alpha) \approx \Phi\left(\frac{U-\mu_1}{\sigma/\sqrt{n}}\right) - \Phi\left(\frac{L-\mu_1}{\sigma/\sqrt{n}}\right).$$

This makes the roles of effect size, variability, sample size and the chosen error rate visible.

What changes power?

- **Larger effect in the tested direction.** The alternative distribution moves farther into the rejection region. For this symmetric two-sided mean test, larger $|\mu_1 - \mu_0|$ means higher power.
- **Larger n.** With the same model, effect, variability and α, the standard error shrinks and the alternative becomes easier to distinguish.
- **Smaller variability.** More precise measurement or a more efficient design can also increase power; collecting more observations is not the only possible design improvement.
- **Larger α.** The rejection region expands, increasing power but also allowing more Type I errors. Choose this tradeoff before examining results.



For a one-sided test, power concerns the chosen direction: a right-sided test is not designed to detect a decrease. The alternative must follow the scientific question, not whichever tail is favorable after observing data.

Statistical significance and scientific importance

A very large sample can detect a tiny, unimportant effect. A small sample can miss an important effect. Report the estimated effect, its uncertainty and its scientific context along with the p-value. Power is most useful for planning detection of an effect that matters scientifically.

Pause and explain. Why can lowering α from .05 to .01 increase the probability of missing a real effect, if everything else is fixed?

Two Independent Means: Derive the Scale

Let $X_{11}, \ldots, X_{1n_1}$ and $X_{21}, \ldots, X_{2n_2}$ be independent random samples from two populations with means μ_1, μ_2 and variances σ_1^2, σ_2^2. The target is the population difference $\Delta = \mu_1 - \mu_2$.

$$H_0 : \Delta = \Delta_0, \qquad H_a : \Delta > \Delta_0,\ \Delta < \Delta_0,\ \text{or } \Delta \neq \Delta_0.$$

Usually $\Delta_0 = 0$, but a specified nonzero difference can be tested with the same formula.

The variance follows from independence

The natural estimate is $\overline{X}_1 - \overline{X}_2$. Its mean and variance are

$$\begin{aligned} \mathrm{E}(\overline{X}_1 - \overline{X}_2) &= \mu_1 - \mu_2, \\ \mathrm{Var}(\overline{X}_1 - \overline{X}_2) &= \mathrm{Var}(\overline{X}_1) + \mathrm{Var}(\overline{X}_2) - 2\,\mathrm{Cov}(\overline{X}_1, \overline{X}_2) \\ &= \frac{\sigma_1^2}{n_1} + \frac{\sigma_2^2}{n_2}. \end{aligned}$$

Independent estimation errors can push the difference in either direction. Subtracting the estimates does not subtract their variances.

Known and estimated standard deviations

With known population SDs,

$$Z = \frac{(\overline{X}_1 - \overline{X}_2) - \Delta_0}{\sqrt{\sigma_1^2/n_1 + \sigma_2^2/n_2}}.$$

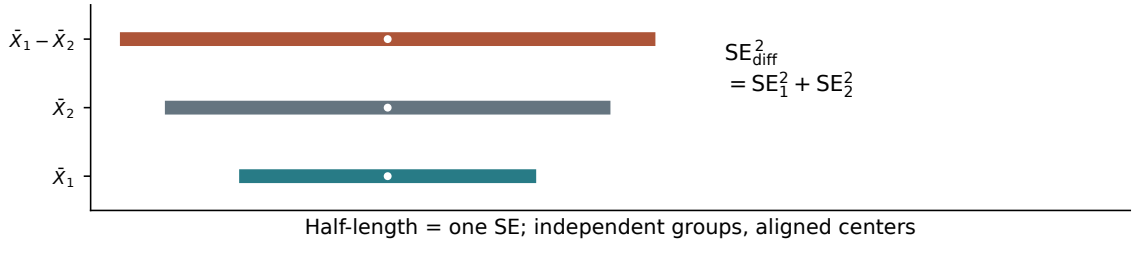
This is standard normal under H_0 for independent normal populations, and approximately standard normal when adequate CLT approximations hold. With unknown SDs and sufficiently large samples in *both* groups, use

$$\boxed{z = \frac{(\overline{x}_1 - \overline{x}_2) - \Delta_0}{\sqrt{s_1^2/n_1 + s_2^2/n_2}}.}$$

Then use the same tail formulas and critical values as on p. 15. The course guideline is $n_1, n_2 \geq 30$, subject to distribution shape and sampling assumptions.

This large-sample method does **not** assume equal population variances and does not pool them. The pooled small-sample t method in Chapter 9 adds an equal-variance assumption. Identify the method from the model, not just from the fact that two groups appear.

Pause and explain. If group 1 is measured much more precisely than group 2, can the uncertainty in the difference ignore group 2? Point to the term that prevents this.



Independent groups: a minus sign in the contrast does not subtract variance.

Comparing GPA at Two Schools

Lecture-slide example: GPA comparison, original slides 44–49 / PDF pp. 57–62.

Question. Independent random samples from two large high schools give the following data. Is there sufficient evidence that School A's population mean GPA is lower than School B's? Use $\alpha = .05$.

	School A	School B
Sample size	36	45
Sample mean GPA	2.24	2.31
Sample variance	0.0560	0.0540

1. Choose an order and keep it

Let $\Delta = \mu_A - \mu_B$. Then

$$H_0 : \Delta = 0, \qquad H_a : \Delta < 0.$$

The samples are stated to be independent and random. Both sample sizes meet the course large-sample guideline. We also assume the GPA distributions permit an adequate normal approximation to their mean difference.

2. Calculate the statistic

The table gives *variances*, so do not square 0.0560 and 0.0540 again:

$$\widehat{\mathrm{SE}}(\overline{X}_A - \overline{X}_B) = \sqrt{\frac{0.0560}{36} + \frac{0.0540}{45}} \approx 0.0524934,$$
$$z = \frac{(2.24 - 2.31) - 0}{0.0524934} \approx -1.3335.$$

The negative sign points toward the chosen alternative. It does not by itself establish significance.

3. Use the left tail and conclude

$$p\text{-value} = \Phi(-1.3335) \approx 0.09118 > 0.05.$$

Equivalently, $-1.3335 > -1.645$, so we do not enter the left rejection tail. **Fail to reject** H_0: the data do not provide sufficient evidence at the 5% level that School A's mean GPA is lower.

The observed difference is -0.07 GPA points. For scale, the associated two-sided 95% interval is approximately

$$-0.07 \pm 1.960(0.0524934) = [-0.1729,\ 0.0329].$$

This interval is a useful uncertainty summary, though a two-sided 95% interval is not the exact inversion of this one-sided 5% test.

Full-precision calculation gives $p \approx .09118$. The slides round the statistic to -1.33 and report .0918. Both lead to the same conclusion.

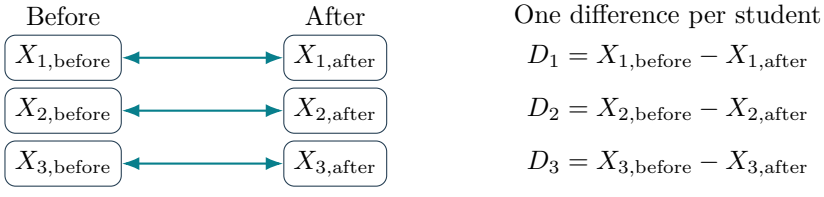
Paired Data: First Form the Differences

Lecture-slide example: Screen-time experiments, original slides 50–52 / PDF pp. 63–65.

The slides contrast two designs:

1. Record one class's grades for six weeks without a screen-time limit, then for six weeks with a one-hour-per-day limit for each student.
2. Compare two different classes of the same grade level: one without a restriction, the other with a one-hour-per-day limit.

The first design measures the *same students twice*. Their observations are naturally paired. The second has distinct groups; it does not provide one-to-one matches.



A paired comparison becomes a one-mean problem

Define $D_i = X_{i,\text{before}} - X_{i,\text{after}}$, with independent pairs across students. Test the population mean difference:

$$H_0 : \mu_D = 0, \qquad H_a : \mu_D \neq 0$$

for the slides' two-sided question. If the research question specifically predicts higher grades after restriction, this ordering would instead use $H_a : \mu_D < 0$.

$$\overline{D} = \frac{1}{n}\sum_{i=1}^{n} D_i, \qquad s_D^2 = \frac{1}{n-1}\sum_{i=1}^{n}(D_i - \overline{D})^2, \qquad z = \frac{\overline{d} - 0}{s_D/\sqrt{n}}.$$

Here n counts **pairs**, not total measurements. Use the large-sample normal approximation when justified; Chapter 9 supplies a small-sample paired t method.

Why the independent-samples formula is wrong for pairs

$$\operatorname{Var}(D) = \operatorname{Var}(X_{\text{before}}) + \operatorname{Var}(X_{\text{after}}) - 2\operatorname{Cov}(X_{\text{before}}, X_{\text{after}}).$$

Computing s_D from the actual differences respects the within-student association. Ignoring the pairing discards that information.

Design limits remain: one before/after class also confounds the restriction with time. Two whole classes may have within-class dependence and class-level differences. The slide example distinguishes pairing; it does not by itself establish independent student-level errors or a causal effect.

One Proportion: The Null Sets the Variance

Let $X_i = 1$ for a success and 0 otherwise. With independent Bernoulli trials of common probability p,

$$X = \sum_{i=1}^{n} X_i \sim \text{Bin}(n, p), \qquad \widehat{p} = \frac{X}{n}, \qquad \text{E}(\widehat{p}) = p, \qquad \text{Var}(\widehat{p}) = \frac{p(1-p)}{n}.$$

Thus a proportion is also a sample mean, but its variance is tied to its mean. Under $H_0 : p = p_0$, let $q_0 = 1 - p_0$. The null reference model is

$$\widehat{p} \overset{\text{approx}}{\sim} N\left(p_0, \sqrt{\frac{p_0 q_0}{n}}\right), \qquad \boxed{Z = \frac{\widehat{p} - p_0}{\sqrt{p_0 q_0 / n}} \overset{\text{approx}}{\sim} N(0,1).}$$

This is the same standardization used in Chapter 7's stipulated-proportion probability calculations. The new ingredient is using the tail probability to test a claim.

Check the conditions before using the normal curve

The course uses $np_0 > 5$ and $nq_0 > 5$ as its large-count guideline. These are expected success and failure counts *under the null*. Also check independence and a representative sampling design. Rare events can require a very large n; the label "large sample" alone is insufficient.

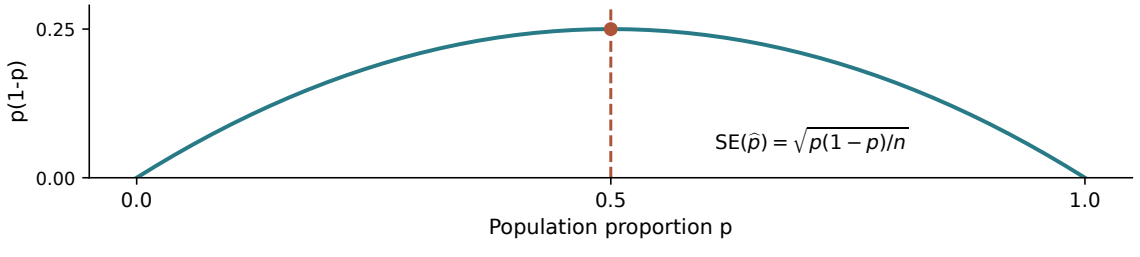
Choose $H_a : p > p_0$, $p < p_0$ or $p \neq p_0$, then use the corresponding normal tail. This chapter's worked examples use the direct normal approximation without a continuity correction. Exact binomial methods are a different calculation when the approximation is unsuitable.

Connect to the Chapter 7 confidence interval

Estimating an unknown p	Testing a specified p_0
Wald interval estimates the variability from data: $\sqrt{\widehat{p}(1-\widehat{p})/n}$.	The null supplies its own variability: $\sqrt{p_0(1-p_0)/n}$.

The roles differ, so the standard errors can differ. Inverting this null-standardized test gives a score interval, not the simple Chapter 7 Wald interval. Therefore "the Wald interval excludes p_0" need not agree exactly with the test at a borderline case.

Pause and explain. Why does changing the null value change the reference standard error in a proportion test, even though the observed data stay fixed?



The null fixes p_0 for a test; a confidence interval estimates variability from $\widehat{p}$.

Defective Chips: A Small Proportion, a Large Sample

Lecture-slide example: Computer-chip example, original slides 55–60 / PDF pp. 68–73; wording condensed.

Question. The current manufacturing process produces defective chips at rate 0.3%. A proposed process yields 15 defective chips in a sample of 10,000. Is there evidence that the new process reduces the defect rate, using a 1% significance level?

The slide's phrase about being "99% sure" is implemented here as $\alpha = .01$. It does not mean that the result will assign a 99% probability to the alternative hypothesis.

1. State the claim and check the null counts

Let p be the new process's defect probability. Then

$$H_0 : p = .003, \qquad H_a : p < .003, \qquad \widehat{p} = \frac{15}{10000} = .0015.$$

Assume independent, representative chips with a stable defect probability. Under H_0,

$$np_0 = 30 > 5, \qquad n(1 - p_0) = 9970 > 5.$$

Both expected counts meet the course guideline even though p_0 itself is small.

2. Standardize using the null probability

$$\mathrm{SE}_0 = \sqrt{\frac{.003(.997)}{10000}} \approx .000546900, \qquad z = \frac{.0015 - .003}{.000546900} \approx -2.74273.$$

The estimate is about 2.74 null standard errors below the old defect rate. For the left-sided alternative,

$$p\text{-value} = \Phi(-2.74273) \approx .003047 < .01.$$

Thus **reject** H_0. The data support a lower defect probability for the proposed process at the 1% significance level, subject to the sampling assumptions. The critical-value route agrees: $-2.74273 < -2.32635$.

3. Report the size of the change

The observed rate decreased from the benchmark 0.30% to 0.15%: a difference of 0.15 *percentage points*, or a 50% reduction relative to the benchmark. The test decision and the practical magnitude are distinct pieces of information.

The slides report approximately $z = -2.73$ and $p = .0032$. Evaluating the displayed data without early rounding gives $z = -2.74273$ and $p = .003047$. The hypothesis-test conclusion is unchanged.

Pause and explain. Would 15 defects out of only 100 chips give the same sample proportion or the same evidence? Explain which quantities would change.

Fitness: A Proportion Test with Strong Evidence

Lecture-slide example: Fitness survey, original slides 61–66 / PDF pp. 74–79.

Question. A large survey concluded that 20% of US adults participate in fitness activities at least once a week. You suspect that the proportion is lower for adults over 40. In a survey of 2,500 US adults over 40, 375 report weekly fitness activity. Test at $\alpha = .05$. Then find the test's power if the actual proportion is .179 (continued on p. 26).

1. Identify what is being treated as fixed

Let p be the weekly-fitness proportion among adults over 40. As the slides do, treat .20 as the specified population benchmark:

$$H_0 : p = .20, \qquad H_a : p < .20.$$

If .20 instead had appreciable estimation uncertainty from another survey, we would need a model accounting for that uncertainty. It is not included in this one-sample calculation.

Assume the survey gives independent, representative Bernoulli observations. The null expected counts are $2500(.20) = 500$ and $2500(.80) = 2000$, so the normal approximation meets the course count guideline.

2. Compute the observed statistic and tail probability

$$\widehat{p} = \frac{375}{2500} = .15, \qquad \mathrm{SE}_0 = \sqrt{\frac{.20(.80)}{2500}} = .008, \qquad z = \frac{.15 - .20}{.008} = -6.25.$$

The sample proportion lies 6.25 null standard errors below the benchmark. Hence

$$p\text{-value} = \Phi(-6.25) \approx 2.05 \times 10^{-10} < .05.$$

Reject H_0. There is strong evidence, within this sampling model, that the weekly-fitness proportion among adults over 40 is below the specified 20% benchmark.

3. Read an extreme result carefully

A table or calculator may display this tail probability as 0.0000. The probability is extremely small, not exactly zero. Nor is it the probability that H_0 is true.

The observed difference is $-.05$, or five percentage points. This survey comparison does not by itself establish that aging *causes* a reduction in exercise; the statistical parameter and the causal claim are different questions.

The next calculation does not reuse the observed .15 as the true alternative. The problem explicitly asks about a hypothetical population with $p = .179$. Power concerns repeated samples from that specified population.

Fitness: Power Uses Two Standard Errors

For the same test, what is the chance of rejection if the true proportion is $p_1 = .179$? Keep the test fixed at $H_0 : p = .20$ versus $H_a : p < .20$, with $n = 2500$ and $\alpha = .05$.

1. Find the rejection cutoff under H_0

$$\frac{\widehat{p} - .20}{.008} \le -1.644854 \iff \widehat{p} \le c = .20 - 1.644854(.008) \approx .186841.$$

The cutoff c belongs to the *rule*. It stays fixed when we evaluate that rule under another population.

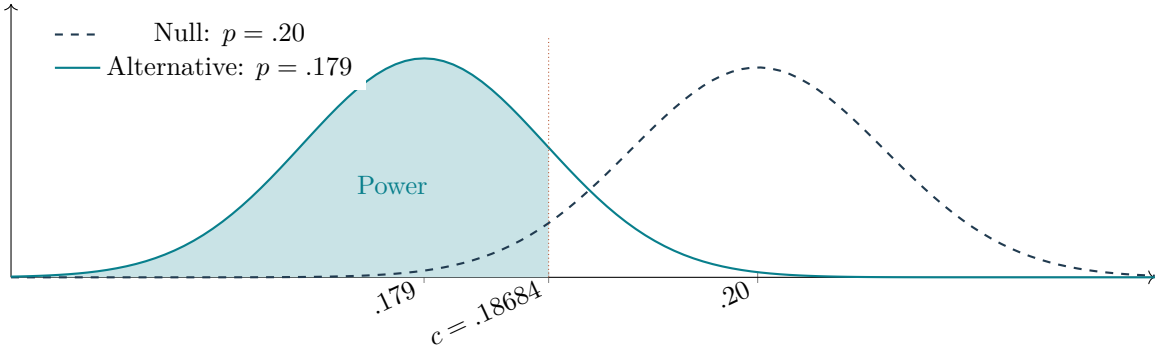
2. Find the sampling distribution under p_1

Under the alternative, both the mean *and* variance of the Bernoulli observations change:

$$\widehat{p} \overset{\text{approx}}{\sim} N\left(.179, \sqrt{\frac{.179(.821)}{2500}}\right) = N(.179, .00766705).$$

Therefore

$$\begin{aligned} \text{Power}(.179) &\approx \Pr(\widehat{p} \le .186841 \mid p = .179) \\ &= \Phi\left(\frac{.186841 - .179}{.00766705}\right) \approx \boxed{.8468}, \qquad \beta \approx .1532. \end{aligned}$$



The shaded alternative area is power; the null area to the left of the same cutoff is approximately α. The normal calculation is continuous, while the binomial sample proportion changes in steps of $1/2500$; this is an approximation, not an exact binomial power calculation.

The slides round $z_{.05}$ to 1.64 and then round the alternative standardized cutoff to 1.03, obtaining .8485. Keeping the normal critical value and arithmetic at full precision gives .8468. Either is about 85%; the approximation and rounding convention should be stated.

Reading Power Curves

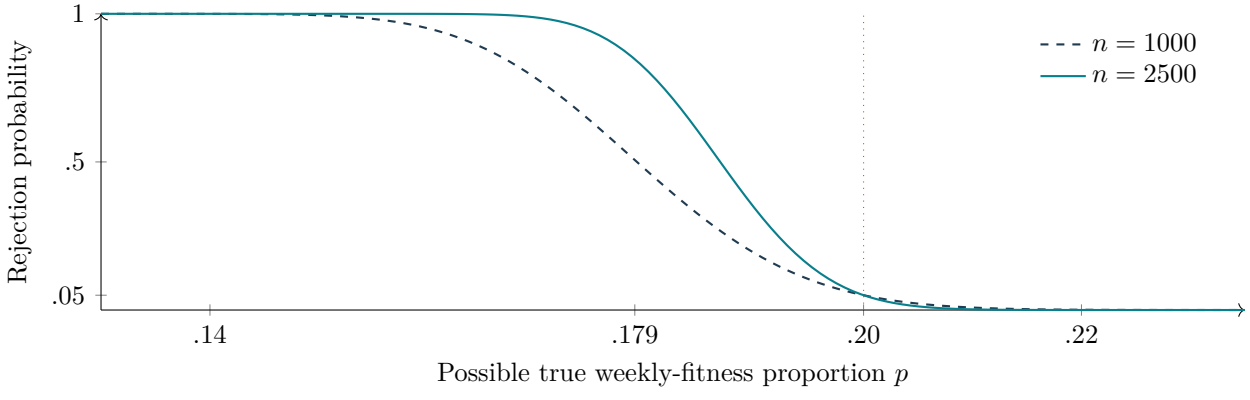
A power curve evaluates the *same test* under many possible true parameter values. For the left-sided fitness test, let

$$c_n = .20 - z_{.05}\sqrt{\frac{.20(.80)}{n}}.$$

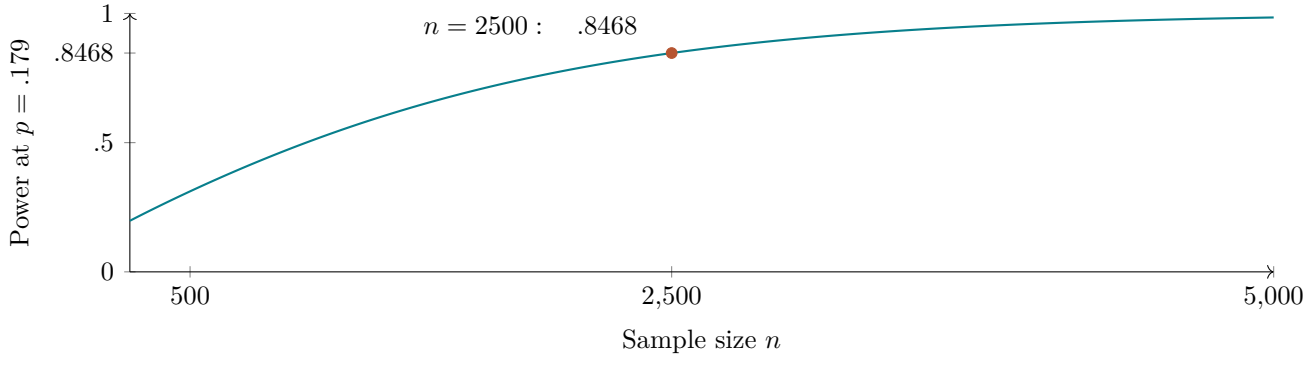
At a possible true proportion p, the normal approximation gives

$$\text{Power}_n(p) \approx \Phi\left(\frac{c_n - p}{\sqrt{p(1-p)/n}}\right).$$

The rejection rule uses .20; the distribution used to evaluate its probability uses the possible true p.



At $p = .20$, the rejection probability is the approximately calibrated Type I error rate .05. For $p < .20$, rejection is a correct detection and its probability is power. Values above .20 point opposite the stated alternative; this left-sided test is not designed to detect them.



The second curve fixes the scientifically relevant departure at $p = .179$ and changes n. Each sample size gets a newly calibrated 5% null cutoff. Increased precision makes this fixed decrease easier to detect. These curves use the slide example's model and the same normal approximation as p. 26.

Two Proportions: Why Pool Under the Null?

Suppose two independent groups supply success counts X_1, X_2 with sizes n_1, n_2 and probabilities p_1, p_2. The observed proportions are $\widehat{p}_1 = X_1/n_1$ and $\widehat{p}_2 = X_2/n_2$.

This chapter tests equality:

$$H_0 : p_1 - p_2 = 0, \qquad H_a : p_1 - p_2 > 0,\ < 0,\ \text{or}\ \neq 0.$$

The null value is zero for the equality tests studied here. That is a scope choice, not a claim that every possible question about two proportions must test a zero difference.

The null imposes a common success probability

Independence gives

$$\operatorname{Var}(\widehat{p}_1 - \widehat{p}_2) = \frac{p_1(1-p_1)}{n_1} + \frac{p_2(1-p_2)}{n_2}.$$

Under H_0, write $p_1 = p_2 = p$. Then

$$\operatorname{Var}_0(\widehat{p}_1 - \widehat{p}_2) = p(1-p)\left(\frac{1}{n_1} + \frac{1}{n_2}\right).$$

The common p is unknown. Since both samples estimate the *same* null probability, estimate it from all trials together:

$$\widehat{p} = \frac{X_1 + X_2}{n_1 + n_2} = \frac{n_1\widehat{p}_1 + n_2\widehat{p}_2}{n_1 + n_2}, \qquad \widehat{q} = 1 - \widehat{p}.$$

This is a sample-size-weighted average. It is not the simple average of the two proportions unless the sample sizes are equal.

The pooled two-proportion statistic

$$\boxed{z = \frac{\widehat{p}_1 - \widehat{p}_2}{\sqrt{\widehat{p}\widehat{q}(1/n_1 + 1/n_2)}}.}$$

Check independent groups and independent representative observations within each group. The slides check more than five observed successes and failures in each group; also inspect the fitted null expected counts $n_i\widehat{p}, n_i(1-\widehat{p})$ for the normal approximation. Sparse counts need a different approach.

For a confidence interval estimating $p_1 - p_2$, equality is *not* assumed. The Chapter 7 standard error is unpooled:

$$\widehat{\mathrm{SE}}_{\mathrm{CI}} = \sqrt{\frac{\widehat{p}_1(1-\widehat{p}_1)}{n_1} + \frac{\widehat{p}_2(1-\widehat{p}_2)}{n_2}}.$$

Pooling is justified by the specific null model, not merely because two proportions appear.

		Variance used (SE is its square root)
Equality-null test	$\widehat{p}_1, \widehat{p}_2 \longrightarrow \widehat{p}$	$\widehat{p}(1-\widehat{p})(1/n_1 + 1/n_2)$
Difference CI	$\widehat{p}_1, \widehat{p}_2$ stay separate	$\widehat{p}_1(1-\widehat{p}_1)/n_1 + \widehat{p}_2(1-\widehat{p}_2)/n_2$

Only the equality null supplies a common probability to pool.

Heart Disease: A Pooled Comparison

Lecture-slide example: Heart Disease, original slides 72–77 / PDF pp. 85–90.

Question. Hospital records show that 52 men in a sample of 1,000 men, versus 23 women in a sample of 1,000 women, were admitted because of heart disease. Is there sufficient evidence of a higher rate of heart-disease admission among men admitted to the hospital? Use $\alpha = .05$.

1. Define the populations and the direction

Let p_1, p_2 be the respective proportions of admitted male and female patients whose admission was for heart disease. Test

$$H_0 : p_1 - p_2 = 0, \qquad H_a : p_1 - p_2 > 0.$$

Assume independent, representative patient samples without repeated-person dependence. Observed success/failure counts are $52, 948$ and $23, 977$, all above five. The fitted common probability is

$$\widehat{p}_1 = .052, \qquad \widehat{p}_2 = .023, \qquad \widehat{p} = \frac{52 + 23}{1000 + 1000} = .0375.$$

The fitted null expected counts in each group are 37.5 successes and 962.5 failures, also comfortably above five.

2. Use the common null variability

$$\widehat{\mathrm{SE}}_0 = \sqrt{.0375(.9625)\left(\frac{1}{1000} + \frac{1}{1000}\right)} \approx .00849632,$$
$$z = \frac{.052 - .023}{.00849632} \approx 3.41324,$$
$$p\text{-value} = 1 - \Phi(3.41324) \approx .000321.$$

Since $.000321 < .05$, **reject** H_0. The data provide evidence of a larger proportion of heart-disease admissions among the male patients in the hospital population represented by these samples.

3. Keep the interpretation within the sampling frame

The estimated difference is .029, or 2.9 percentage points. These data concern people *admitted to this hospital*. They do not directly estimate heart-disease prevalence among all men and women, and the comparison does not by itself identify a causal effect.

Using the displayed counts gives $z \approx 3.41324$. The slides display 3.4118; both round to a right-tail probability of about .0003 and give the same conclusion.

Pause and explain. If the two sample sizes were unequal, would taking $(\widehat{p}_1 + \widehat{p}_2)/2$ generally estimate the common null probability using every trial equally? Explain.

Choose the Model Before the Formula

Target/design	Estimate	Chapter 8 test standard error
One mean	$\overline{x}$	$\sigma/\sqrt{n}$ known; otherwise $s/\sqrt{n}$ for the large-sample approximation
Two independent means	$\overline{x}_1 - \overline{x}_2$	$\sqrt{\sigma_1^2/n_1 + \sigma_2^2/n_2}$; substitute s_i when both samples justify the approximation
Paired mean change	$\overline{d}$	$s_D/\sqrt{n}$, with n pairs
One proportion	$\widehat{p}$	$\sqrt{p_0(1-p_0)/n}$ from the null
Two proportions, equality null	$\widehat{p}_1 - \widehat{p}_2$	$\sqrt{\widehat{p}(1-\widehat{p})(1/n_1 + 1/n_2)}$, pooled under H_0

Four questions before calculating

1. **What is measured?** A numerical response suggests a mean; a binary response suggests a proportion. A count can require a separate model for its variance.
2. **How are observations related?** One sample, independent groups and paired measurements have different sources of variation.
3. **What does H_0 specify?** It fixes a parameter value, sometimes also a variance relationship. Use that information when constructing the null reference.
4. **Is the reference distribution justified?** State independence, sampling scope, normality or adequate large-sample conditions. Choose the tail from the research question.

Confidence intervals and tests share the same foundations

Both compare an estimate with sampling variability. An interval reports a range of compatible parameter values; a test evaluates a specified value using a defined rejection rule. For the same fixed-standard-error normal model, a two-sided level-α test and a $100(1-\alpha)\%$ interval agree by algebra. If the procedures use different estimated variances, that exact agreement is not guaranteed.

A one-sided level-α test corresponds to a one-sided $100(1-\alpha)\%$ confidence bound using the same model. It is not generally equivalent to asking whether a two-sided $100(1-\alpha)\%$ interval contains the null value.

Where Chapter 9 begins

If a normal population has unknown variance and the sample is small, estimating the standard error adds uncertainty that the z reference ignores. Chapter 9 develops t reference distributions, degrees of freedom and the corresponding one-sample, independent-sample and paired methods. Small sample size does not turn a Bernoulli proportion into a normal-mean t problem.

Extension: Let the Model Supply the Variance

Additional connection, not a worked example from the Chapter 8 slide deck. The same reasoning applies when a count model relates a mean to its variance. This connects the Chapter 5 Poisson model to Chapter 8 testing without introducing a new testing principle.

A Poisson mean illustrates the pattern

Suppose $X_1, \ldots, X_n$ are independent counts over equal exposure periods, each with distribution Pois(λ). Then

$$\mathrm{E}(X_i) = \lambda, \qquad \mathrm{Var}(X_i) = \lambda, \qquad \mathrm{E}(\overline{X}) = \lambda, \qquad \mathrm{Var}(\overline{X}) = \frac{\lambda}{n}.$$

If $H_0 : \lambda = \lambda_0$ with $\lambda_0 > 0$, the null therefore supplies

$$\mathrm{SE}_0(\overline{X}) = \sqrt{\frac{\lambda_0}{n}}, \qquad Z = \frac{\overline{X} - \lambda_0}{\sqrt{\lambda_0/n}} \overset{\text{approx}}{\sim} N(0,1)$$

when the normal approximation is adequate. Equivalently, the total $T = \sum_i X_i$ has null model Pois($n\lambda_0$) and

$$Z = \frac{T - n\lambda_0}{\sqrt{n\lambda_0}}.$$

The observed mean estimates λ; the hypothesized value supplies the null variance. This is parallel to the Bernoulli null variance $p_0(1 - p_0)$.

The model is doing real work

The equality of Poisson mean and variance is an assumption about the count-generating process. Unequal exposures, clustering, changing event rates or extra variability can invalidate this standard error. A very large sample does not automatically repair a wrong variance model.

A three-line power template

For a left-sided test with rejection cutoff $\overline{X} \le c$:

1. Find c using the null mean and null standard error at the chosen α.
2. Specify a true alternative λ_1 and its distribution, including $\mathrm{SE}_1 = \sqrt{\lambda_1/n}$ in this Poisson model.
3. Evaluate $\Pr_{\lambda_1}(\overline{X} \le c) \approx \Phi((c - \lambda_1)/\,\mathrm{SE}_1)$.

This is exactly the logic used for the fitness proportion: *the rule stays fixed; the distribution used to evaluate its performance changes.*

No homework data are solved here. The purpose is to recognize how a familiar probability model determines the standard error and, through it, the test.

Check Your Understanding

Try these before reading the brief explanations below.

1. At $\alpha = .05$, compare $z = 1.8$ for right-sided and two-sided alternatives.
2. Explain a Type II error for the carpenter example with true mean 20. Can the observed sample alone establish that an error occurred?
3. Why does decreasing α generally reduce power at a fixed true alternative, with sample size and model unchanged?
4. In a proportion test, why does the denominator use p_0? In a power calculation, why may the alternative standard error differ?
5. A researcher measures the same 12 people before and after treatment. What is the basic observational unit for a paired test? Does "24 measurements" justify a two-independent-sample z test?
6. In the heart-disease example, what population does the conclusion concern? What extra claim is not established?

Brief explanations

1. **Right-sided: reject.** $1.8 > 1.645$, and $1 - \Phi(1.8) \approx .0359$. **Two-sided: fail to reject.** $1.8 < 1.960$, and the two-sided p-value is about .0719. The scientific question determines which comparison is appropriate.
2. **Fail to reject $\mu = 19$ when $\mu = 20$.** We cannot know from the sample alone that this error occurred, because the true population mean is unknown. We can calculate the procedure's error probability under a specified model.
3. **The rejection region shrinks.** It contains less probability under the null and also less under the fixed alternative. A lower false-alarm rate can increase the chance of missing a real effect.
4. **The null model sets the null variance.** A Bernoulli variable has variance $p(1 - p)$, so changing its assumed population probability changes the sampling variance. Power uses the alternative's variance to evaluate a cutoff that was set under the null.
5. **Twelve paired differences.** The measurements within a person are related. Analyze the differences, with $n = 12$; do not manufacture independence by counting the measurements separately. Small-sample normal-difference inference leads to the paired t method in Chapter 9.
6. **The sampled hospital-admission population.** The observed proportions concern reasons for admission among admitted patients. They do not directly establish population-wide disease prevalence or causation.

End point

You should now be able to explain a hypothesis test as a sampling-model calculation, justify its standard error and tail, state its conclusion carefully, and evaluate its power at a specified alternative. The next chapter changes the reference distributions when small-sample estimation of variability matters; the underlying logic remains.

Source: course Chapter 8 lecture deck, *Large Sample Hypothesis Testing*, 90-page PDF downloaded November 1, 2026 (77 content frames). The continuation covers original frames 21–77; diagrams and explanations are redrawn for these notes. Worked examples preserve slide data; displayed calculations use full precision unless stated. Page 31 is an explicitly added model-based connection.